

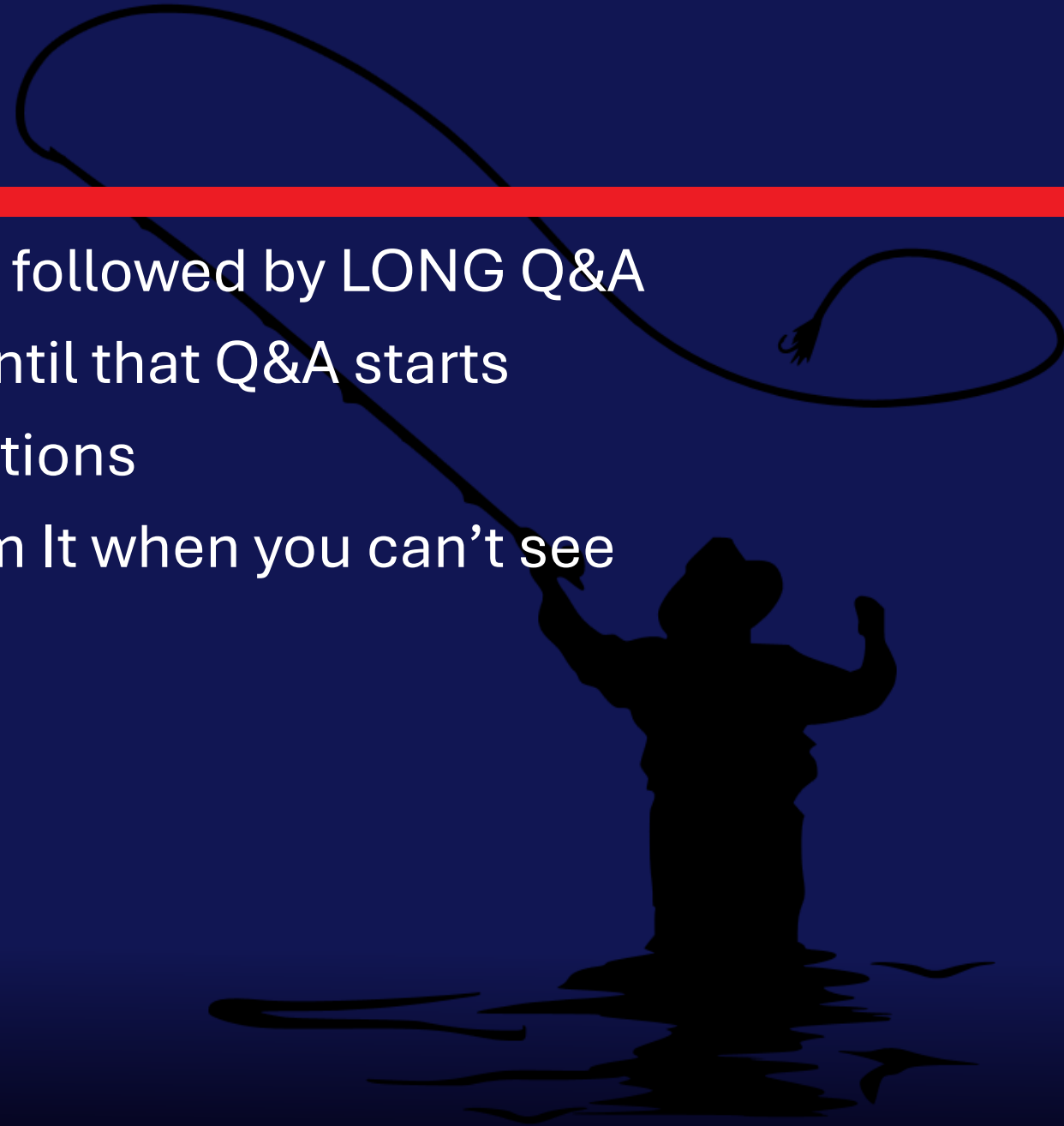
Building Agentic Apps for the IT Pro Admin

From Zero to Agentic — A Hands-On Session
Intune 007 - *Your license to manage.*



Attention

- ◆ MMS sessions are 60-75 minutes followed by LONG Q&A
- ◆ Please hold detailed questions until that Q&A starts
- ◆ Feel free to ask clarification questions
- ◆ Remind the speakers to use Zoom It when you can't see



Speakers



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Senior CE Product Manager



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In/getmodern

Can't see our slides? Can't hear? Need to repeat the question? Call us out!

Session Agenda

What is Vibe Coding

What are we building

How does this work

What do we need to build this?

Permissions and Azure Resources

Building Time!!

Do's and Don't's



The Product Backlog Abyss



Customer Provides Feedback

Alex write a spec

Alex create a work item in ADO

Meeting to discuss the work item

SWE-Lead thinks it's a great idea

SWE-Lead plans it for sprint 4 🙄 🙄 of 2026

The idea dies

The idea was good, but the cost of experimentation was too high.

Vibe Coding ?!?

Vibe coding = describing WHAT you want, and AI writes the code for you

You don't need to know TypeScript, React, or Node.js

You describe intent in plain English — GitHub Copilot generates the code

You review, accept, and run — the AI handles the syntax (There is also Autopilot available)

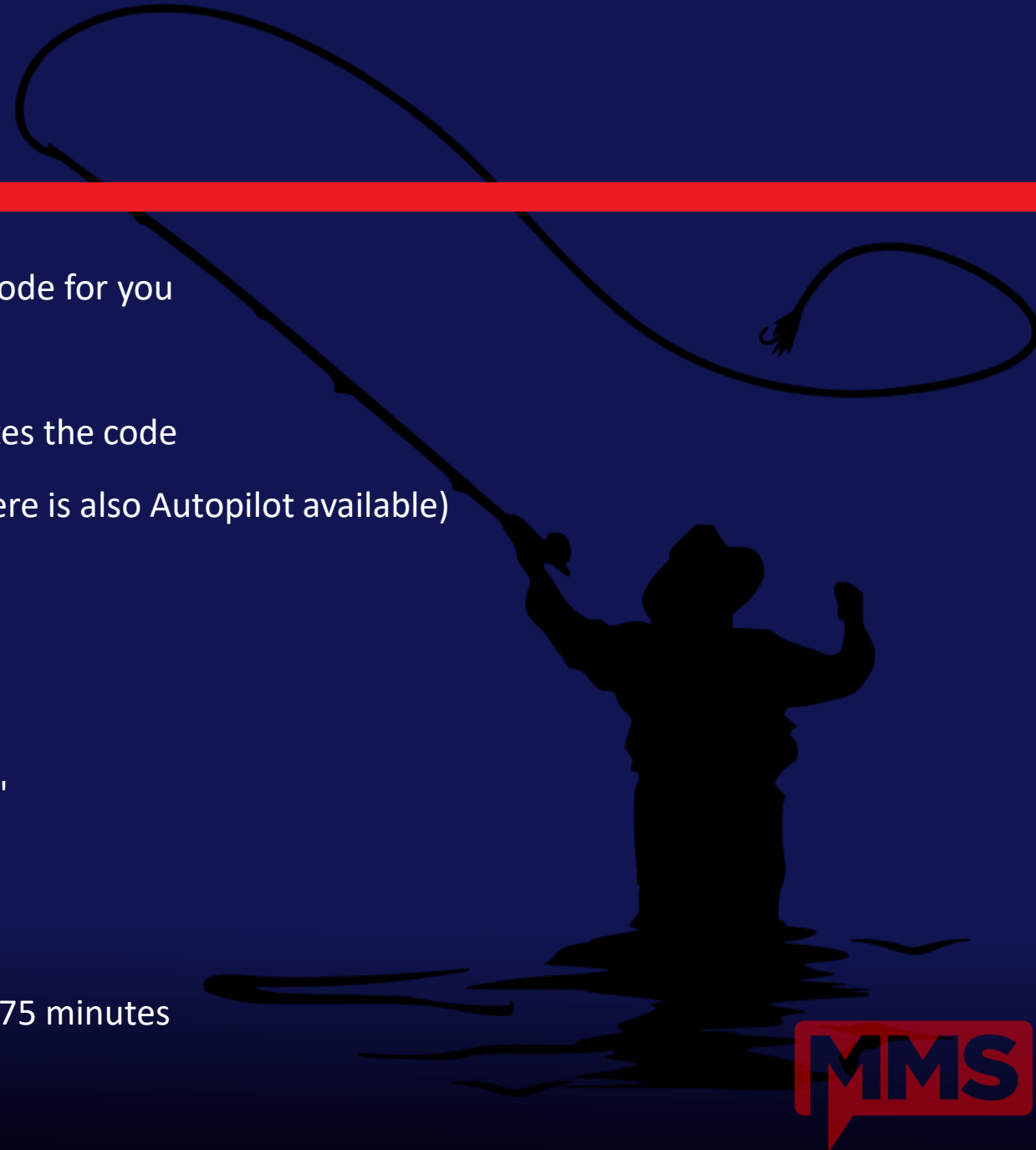
Think of it like giving instructions to a very smart intern:

"Create a server that connects to Microsoft Graph API"

"Build a chat interface that sends messages to the backend"

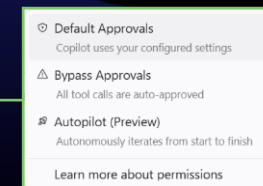
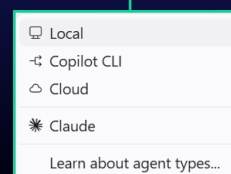
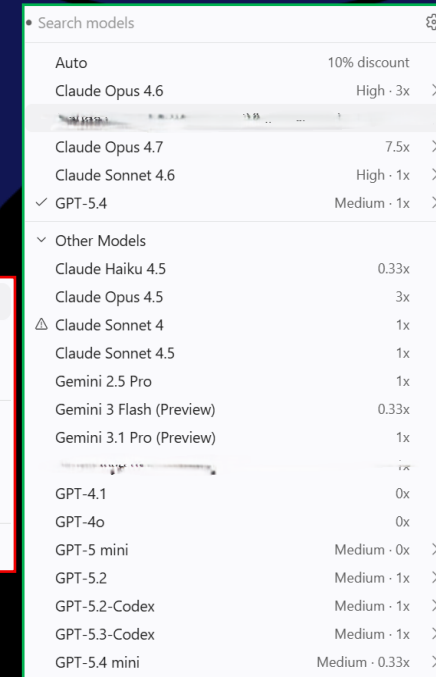
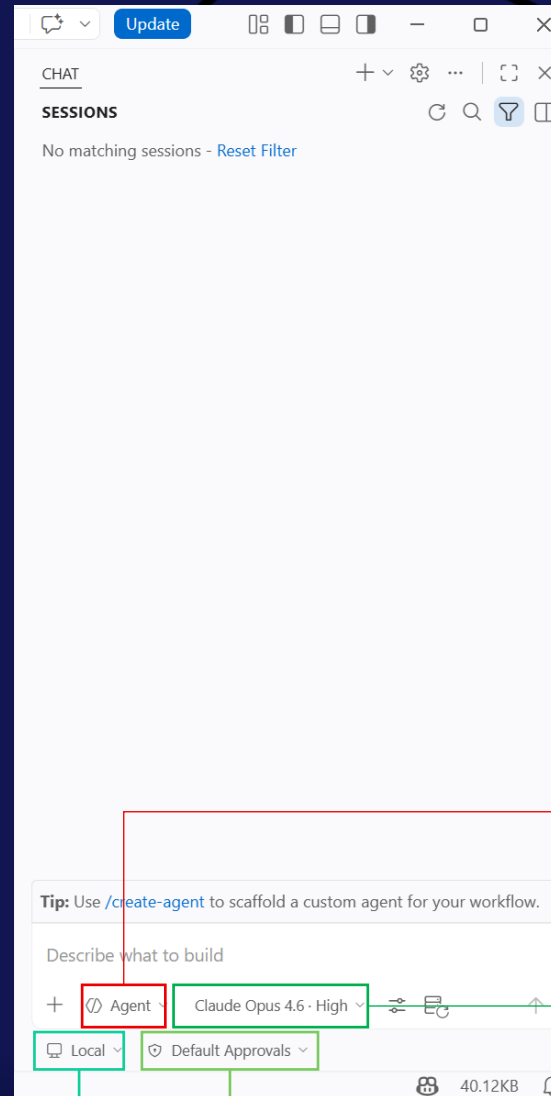
"Add a tool that lists all non-compliant devices"

Today: we'll vibe-code a working Intune management tool in 75 minutes



Let's explorer the interface

- Mode Selector (Ask, Edit, Agent)
- Model Selector
- Input Options (Tools, Extensions, Voice)
- Additional Context (Docs, References)
- Current File



What are we building?

An AI chatbot that manages Microsoft Intune through natural language.

You type: "Show me all non-compliant Windows devices"

The AI: queries the Microsoft Graph API and shows results in a table

MVP Features we'll build:

- ✓ Chat interface (type questions, get answers)
- ✓ Device queries (list devices, filter by OS/compliance/name)
- ✓ Compliance policies (view all policies and their status)
- ✓ Device actions (sync a device, restart a device)
- ✓ Alerts dashboard (automated health monitoring)

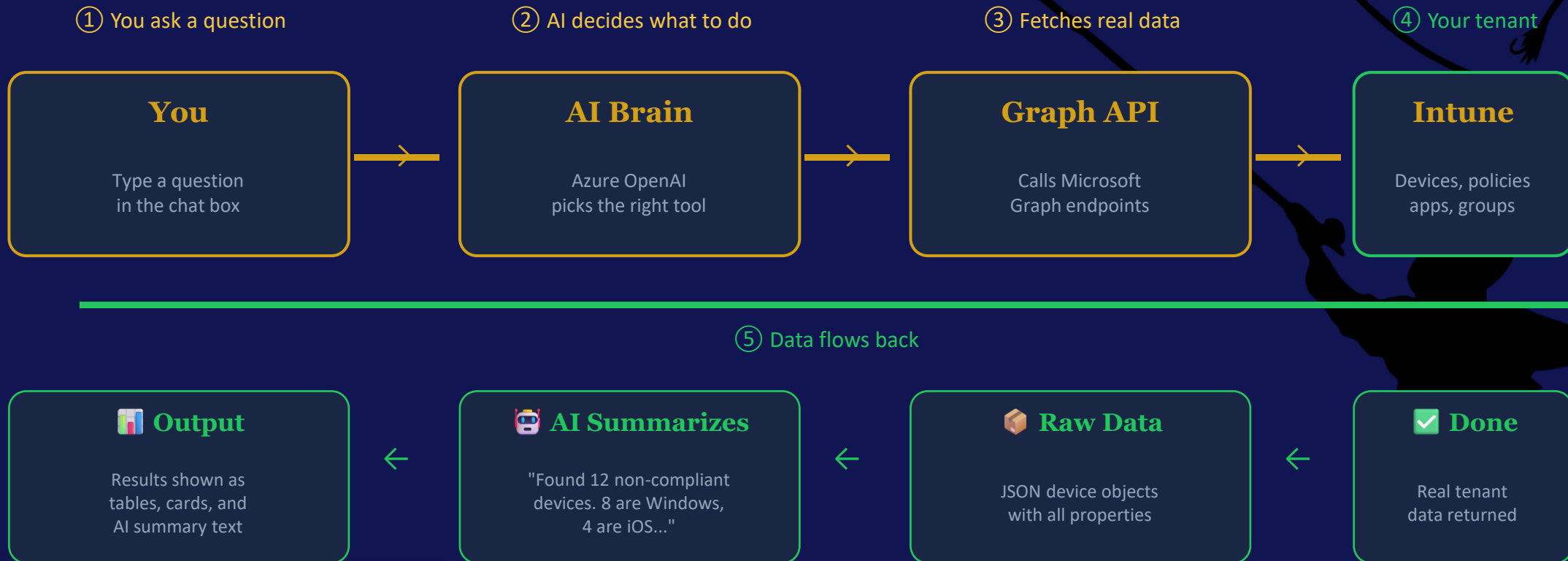
All in 75 minutes. No prior coding experience needed



Pull the prototype showed during the session

[KenGoossens/Intune007 AlphaV1](#)

How it works – The Big Picture



Request → goes right | ← Response comes back left

Prerequisites Checklist

On Your Laptop

- ✓ VS Code installed (free: code.visualstudio.com)
- ✓ GitHub Copilot extension (paid, or free trial)
- ✓ Node.js 18+ installed (nodejs.org)

In Azure

- ✓ An Azure subscription
- ✓ Azure OpenAI resource with a GPT model deployed
- ✓ Entra ID App Registration with Intune permissions
- ✓ Client ID, Client Secret, Tenant ID noted down
(I'll show the permissions on the next slide)

Required Entra ID Permissions

Your app registration needs these Application permissions (with admin consent):

DeviceManagementManagedDevices.ReadWrite.All

DeviceManagementConfiguration.ReadWrite.All

DeviceManagementApps.Read.All

Directory.Read.All

Group.Read.All

Go to: Azure Portal → Entra ID → App registrations → Your app → API permissions

Click "Grant admin consent" after adding all permissions

⚠ Without admin consent, the tool won't be able to access Intune data



New possibilities

Genuinely New / Doesn't Exist Anywhere

Feature	Why it's unique
Self-improving learning engine	No Intune tool learns from user interactions. The few-shot exemplar bank, tool chain pattern learning, and correction memory create an agent that gets better at YOUR tenant over time. Microsoft Copilot in Intune doesn't do this — it's stateless per session.
Automated app icon search + upload	Nobody does this. The 7-source icon search → sharp PNG conversion → Graph API upload pipeline is completely new. In the Intune portal you manually upload icons one by one. The "fix all missing icons" bulk operation doesn't exist anywhere.
Compliance forecast (what-if)	"If I require BitLocker encryption, how many devices would fail?" — this predictive analysis before policy deployment doesn't exist in Intune, Defender, or any third-party tool.
Bulk app rename (find-and-replace)	"Replace 'M365 Labs' with 'Goosnet' in all app names" — there's no tool that does this. In the portal it's one-by-one manual edits.
Benchmark-aware policy generation	Describing a policy in plain English and getting valid Intune JSON with specific CIS L1/L2, NIST, ISO 27001, HIPAA, Essential Eight, Zero Trust, or STIG overlays. Copilot in Intune can help with policies but doesn't have benchmark-specific generation with exact Graph API property names.
Autopilot hash collection via Proactive Remediation	Deploying a deviceHealthScript that collects hardware hashes from already-enrolled devices, outputs them with markers, then auto-imports into Autopilot. This convert-existing-device-to-Autopilot automation chain is new.
PowerShell safety scanning before deployment	15 dangerous pattern checks (Invoke-Expression, credential theft, persistence mechanisms) before deploying AI-generated scripts to Intune. No tool scans remediation scripts for security issues before deployment.
Cross-panel auto-execution	Clicking a device name in an alert detail → panel switches → troubleshooter auto-runs SSE diagnostics → AI root cause analysis. This interconnected drill-down with automatic execution doesn't exist in any Intune management tool.

RAG Pipeline (optimize token consumption)

The RAG pipeline enriches the agent's responses with up-to-date Intune documentation from learn.microsoft.com. When a user sends a message, the pipeline retrieves the most relevant documentation chunks and injects them into the system prompt, giving the LLM factual context alongside its built-in knowledge.

1. Ingest (Scraping)
2. Chunking
3. Embedding
4. Storage
5. Search

Scaling Technique (Avoid API Limits)

All 4 scaling layers implemented and compiling cleanly:

Layer 1 — Device Cache (`deviceCache.ts`):

- SQLite-backed local inventory of all managed devices
- Full sync on first boot, then delta queries every 2 minutes
- OData-to-SQL translation (eq, ne, contains, startsWith, and/or)
- `getManagedDevices()` now reads from cache — zero API calls

Layer 2 — Rate Queue (`rateQueue.ts`):

- 4-tier priority system (IMMEDIATE → AGENT → BACKGROUND → BULK)
- Sliding-window rate budget tracking (9K/10min, 10% buffer)
- 429 detection with Retry-After + exponential backoff + jitter
- P0 requests bypass the queue entirely

Layer 3 — Batch API (`batch.ts`):

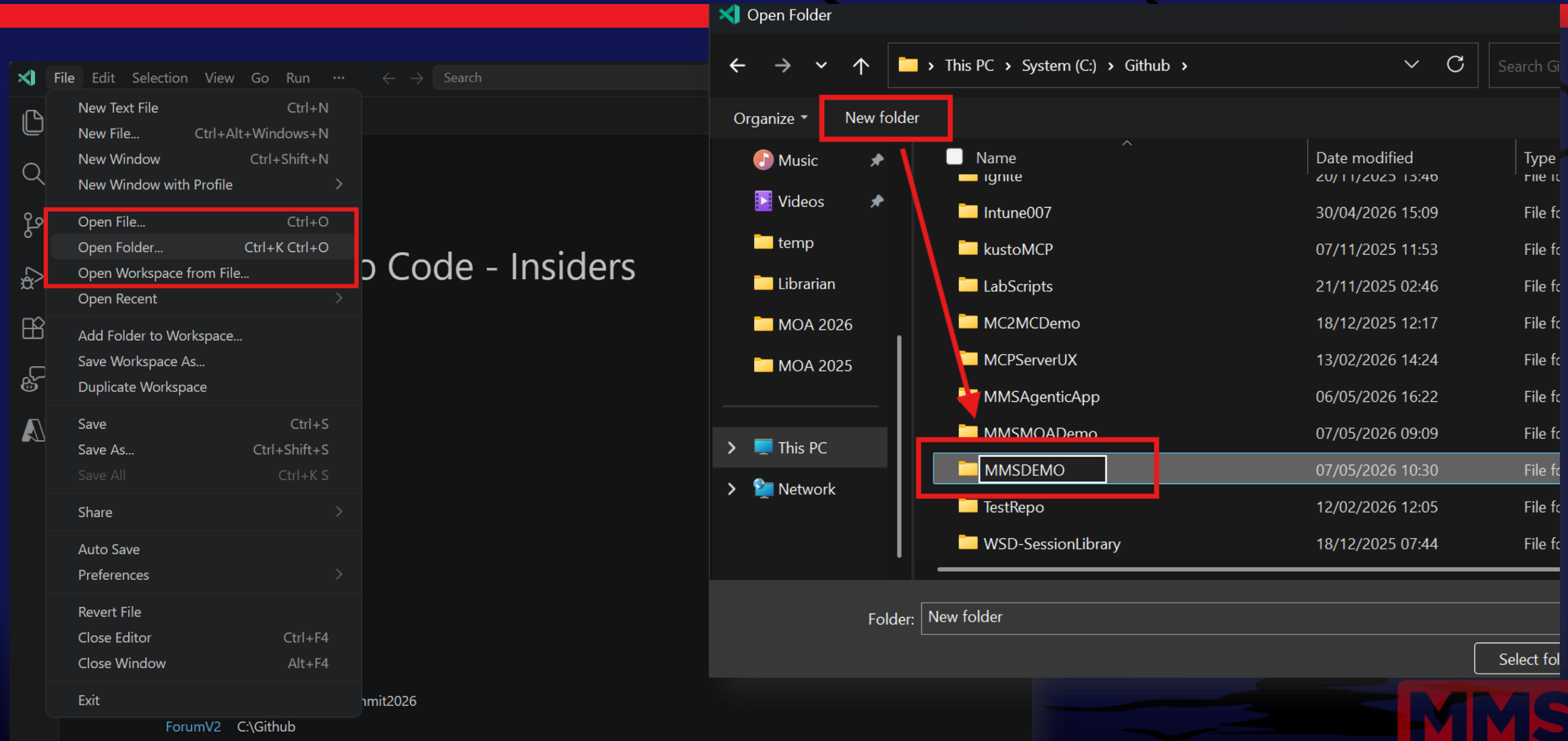
- `executeBatch()` — up to 20 requests per HTTP call, auto-chunks larger batches
- `batchGetDetectedApps()` — 100K devices in 5K batch calls instead of 100K individual calls
- App Health route now uses batch API

Layer 4 — Cache-First Reads (`devices.ts`):

- `getManagedDevices()` reads from SQLite cache when warm
- `getDeviceDetails()` tries cache first, falls back to live API
- `forceRefreshDevice()` for post-action data (sync, restart)
- Transparent — all existing tools work unchanged

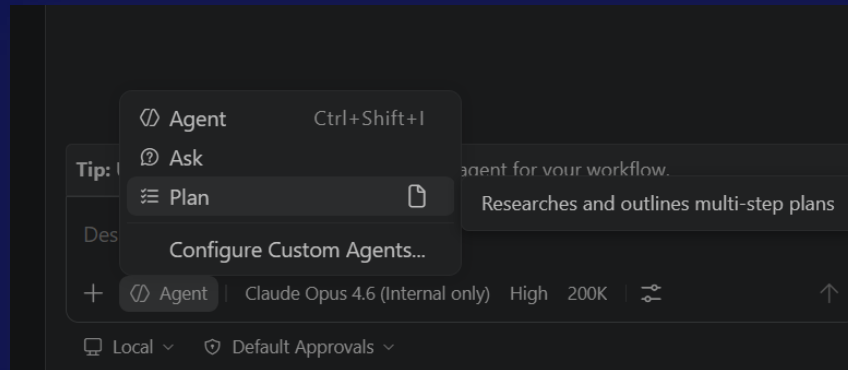


Create the workspace

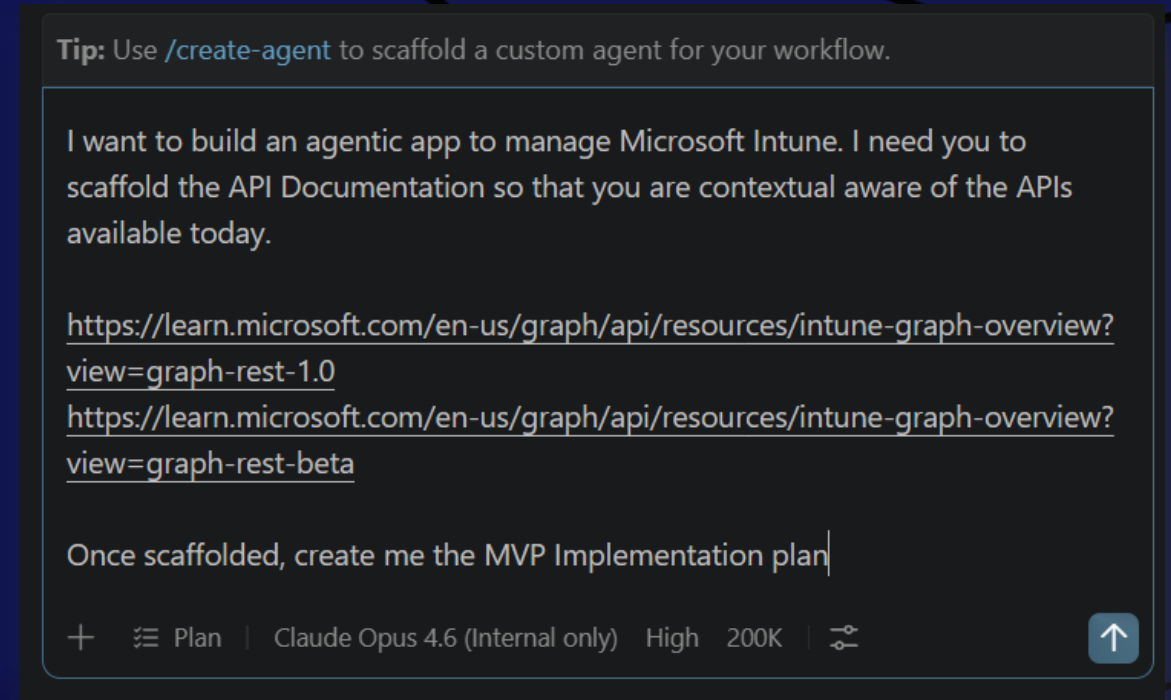


Start planning your project

Change your agent setting to Plan



Start planning by providing the information you want the agent to use.



Let your creativity flow.... Think and Build

007 Intune007
Your license to manage.

CORE

Alerts 1

Device Card

Report Generator

OPERATIONS

Logs

Policies

Policy Builder

Policy Diff

Remediation

INSIGHTS

Analytics

App Health

Autopilot

CVE Monitor

Compliance Forecast

Risk Scores

Security Posture

Tasks

Device Timeline

Settings

Agent

v1.0 · AI-Powered

Device Card

NB-DUYGU-AI1

NB-DUYGU-AI1

Windows

10.0.26200.5733

arM64

compliant

Encrypted

4d ago

Sync

Restart

Troubleshoot

Timeline

Prepare Autopilot

200

Apps

3

Profiles

✓

Compliance

16

RAM (GB)

Profiles: ✓ 3 295GB free / 475GB company

Duygu Goossens

duygu@goosnet.xyz

company

Hardware & Storage

ManufacturerMicrosoft Corporation

ModelMicrosoft Surface Pro, 11th Edition

Chassislaptop

ArchitecturearM64

RAM16 GB

OS EditionEnterprise

180 GB used295 GB free of 475 GB

Security

EncryptionEncrypted

DFCI ManagedNo

Network

Enrollment & Management

007 Intune007 Agent
Your license to manage.

- MMS MOA 2026 Edge Mac OS
- MMS MOA 2026 Excel
- MMS MOA 2026 Managed Browser
- MMS MOA 2026 Microsoft 365 Apps for macOS
- MMS MOA 2026 OneDrive

Before I execute the bulk rename, please confirm:

- Rename all 6 apps exactly like this?
- Replace text "I am Cool -" with "MMS MOA 2026 " (with the space)?

Once you confirm, I'll apply the change across the tenant.

I confirm, make the changes

All done. ✓

6 apps were renamed successfully.

Examples:

- I am Cool - Device Tunnel AOVPN → MMS MOA 2026 Device Tunnel AOVPN
- I am Cool - Edge Mac OS → MMS MOA 2026 Edge Mac OS
- I am Cool - Excel → MMS MOA 2026 Excel
- I am Cool - Managed Browser → MMS MOA 2026 Managed Browser
- I am Cool - Microsoft 365 Apps for macOS → MMS MOA 2026 Microsoft 365 Apps for macOS
- I am Cool - OneDrive → MMS MOA 2026 OneDrive

If you want, I can also:

- enforce a naming standard for all apps in the tenant
- detect apps with inconsistent prefixes
- bulk fix missing icons or publishers for cleaner Company Portal presentation.

Ask about your Intune environment...

The MMS logo is located in the bottom right corner of the image. It consists of the letters "MMS" in a bold, white, sans-serif font, set against a red speech bubble background.

The Rules of Vibe Coding

Be the Orchestrator

- **Do: Define Roles.** Treat Copilot like a team. Explicitly tag @SecurityAgent for audits or @DesignAgent for UI polish.
- **Do: Curate Context.** Use .github/copilot-instructions.md to set the "Constitution" of your project (style guides, tech stack preferences).
- **Do: Iterate on "The What".** Focus your energy on refining the user story and the acceptance criteria.
- **Do: Trust but Verify.** Use the Agent to write the tests *before* you blindly accept the code.

Don't be the Dictionary

- **Don't: Ask the Generalist for Everything.** Don't expect a single prompt to solve architecture, CSS, *and* security in one go.
- **Don't: Repeat Yourself.** Don't type "Use Tailwind CSS and TypeScript" in every single prompt. Put it in the instructions file.
- **Don't: Fixate on "The How".** Don't waste 20 minutes debugging a syntax error. Delete the block and ask the Agent to rewrite it.
- **Don't: Deploy the Vibe.** Don't ship the raw Spark prototype without the "Production Layer" (Security, Logging, Error Handling).

SAVE THE DATES

Oct 25-28, 2026



May 2-6, 2027



Oct 10-13, 2027



Extended Q&A



Recast



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